IPSCE

Inventory Planning and Stock Control Evironment

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Department of CSE

IPSCE

Inventory Planning and Stock Control Environment

ABSTRACT

IPSCES – Inventory Planning and Stock Control Environment System

➤ Application designed to manage inventory operations and financial transactions in real time

➤ Integrates customer management, product tracking, sales, purchases, and expense monitoring

➤ Built using Java Spring Boot backend and dynamic HTML, CSS, and JavaScript frontend

➤ Implements RESTful APIs for seamless communication between frontend dashboard and backend services

➤ Provides invoice generation, financial analytics, and data storage using LocalStorage and database integration

➤ Demonstrates practical implementation of Data Structures and Algorithms concepts in a real-world business system

➤ Offers a centralized dashboard to improve inventory control, financial tracking, and business decision-making

Existing System

Challenges in Traditional Inventory and Business Management:

- ➤ Manual recording of customer, product, and sales data leading to errors and delays
- ➤ Lack of real-time tracking for inventory levels and financial transactions
- ➤ Disconnected management of customers, products, sales, purchases, and expenses
- ➤ Delayed identification of stock shortages or financial inconsistencies
- ➤ No centralized dashboard for monitoring business performance and activities
- ➤ Paper-based or spreadsheet-dependent systems prone to data loss and human errors

Proposed System

IPSCES – Intelligent Inventory and Business Management Solution:

➤ Real-time web-based dashboard with dynamic page navigation and responsive user interface
➤ Java Spring Boot backend integrated with RESTful APIs for efficient data processing
➤ Centralized dashboard displaying customers, products, sales, purchases, and expense information
➤ Automated invoice generation and financial tracking for better business management
➤ Secure user profile management with authentication and basic validation features
➤ DSA Integration: sorting algorithms, searching techniques, arrays, and list-based data handling
➤ Modular system architecture allowing scalability and future feature enhancements

System Architecture

Frontend Layer (Modular SPA

Communication Layer

Backend Layer (Java Spring Boot)

System Architecture

Frontend Layer (Modular SPA)

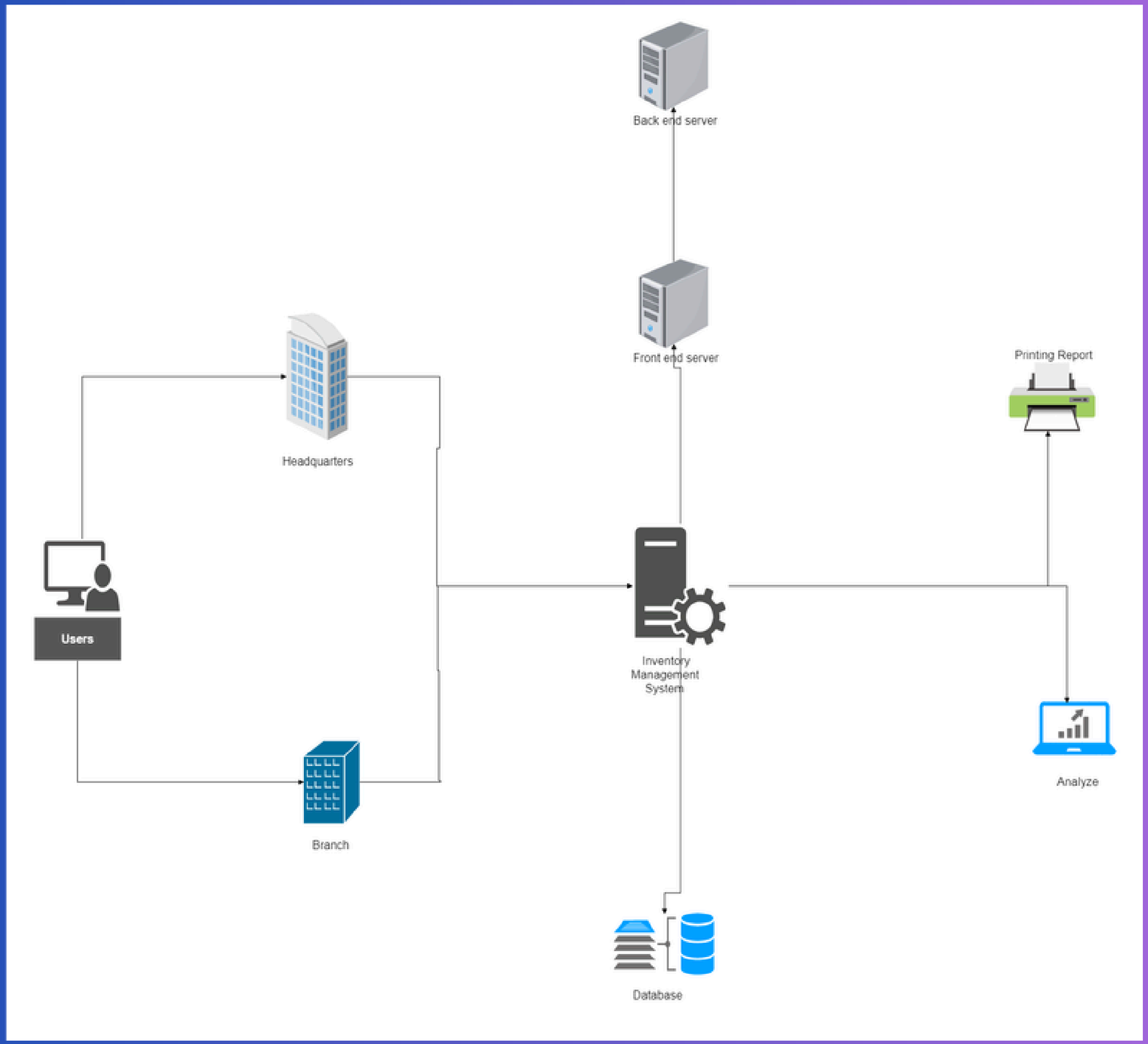
- Developed using HTML, CSS, and JavaScript to provide an interactive dashboard interface
- Allows users to manage customers, products, sales, purchases, expenses, and invoices

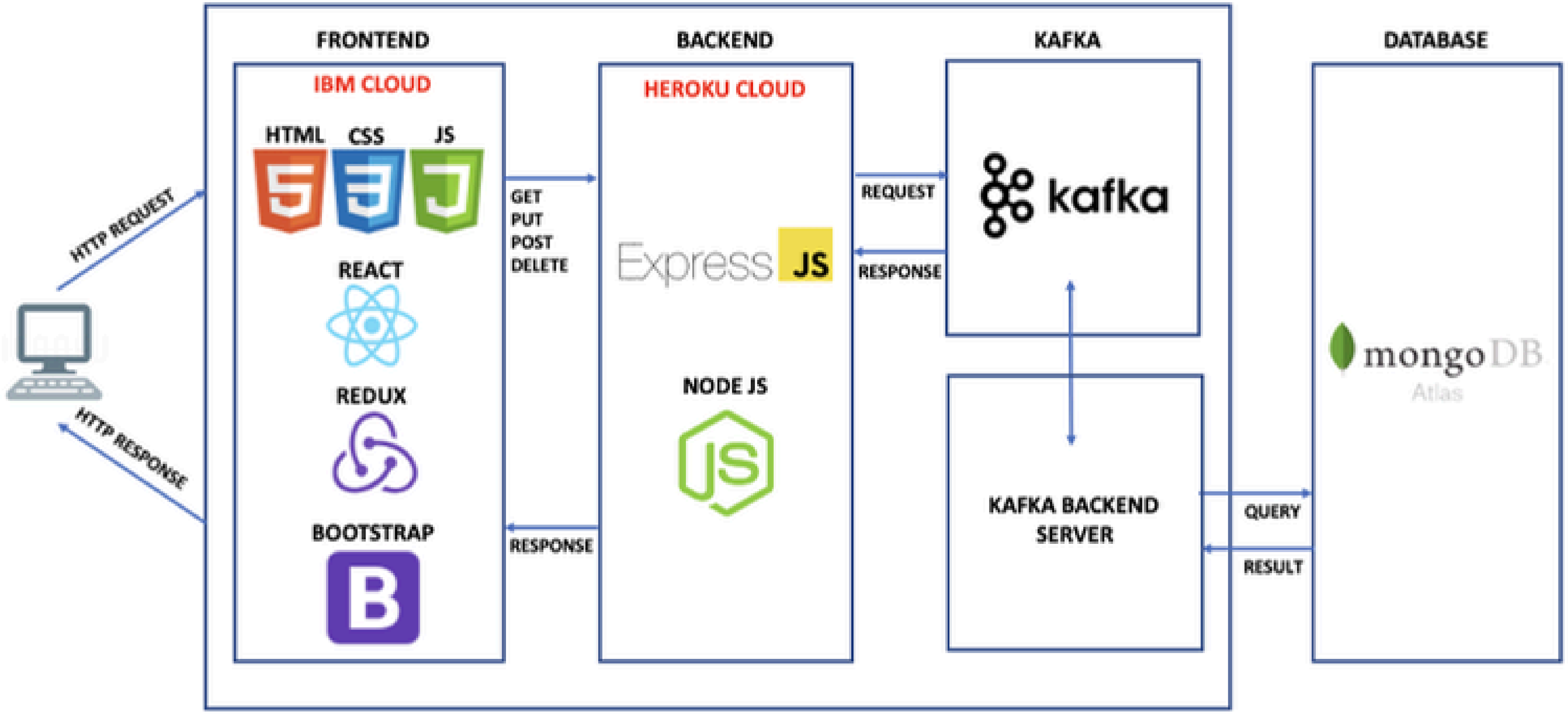
Communication Layer

- Uses RESTful APIs and HTTP requests (Fetch API) for communication between frontend and backend
- Transfers data using JSON format for efficient processing

Backend Layer (Java Spring Boot)

- Implements business logic, API services, and data processing
- Handles inventory management, financial records, and database operations


Back end server
Front end server
Printing Report
Headquarters
Users
Inventory Management System
Analyze
Branch
Database


FRONTEND
IBM CLOUD
HTML
CSS
JS
REACT
REDUX
BOOTSTRAP
BACKEND
HEROKU CLOUD
Express JS
NODE JS
KAFKA
kafka
KAFKA BACKEND SERVER
DATABASE
mongoDB
Atlas
HTTP REQUEST
HTTP RESPONSE
GET
PUT
POST
DELETE
RESPONSE
REQUEST
RESPONSE
QUERY
RESULT

Project Modules

Dashboard Overview

- Displays real-time business information through a centralized dashboard
- Shows statistics for customers, products, sales, purchases, expenses, and profit
- Provides quick insights for monitoring overall business performance

Product Management

Allows users to add, update, and delete products

Tracks product price, availability, and product details

Maintains product records for inventory and billing operations

Customer Management

Stores and manages customer details such as name, email, and contact information

Helps track customer transactions and purchase history

Enables efficient customer relationship management

Sales Management

- Records sales transactions and customer purchases
- Calculates total sales and updates financial summaries
- Provides reports for monitoring business revenue

Purchase Management

Tracks supplier purchases and inventory procurement

Records purchase amounts and supplier details

Helps manage stock availability and inventory updates

Expense Management

Maintains records of business expenses and operational costs

➤ Calculates total expenses for financial analysis

➤ Helps determine overall profit and loss

Invoice Management

Generates professional invoices for customer transactions

Allows exporting invoices and tracking payment status

Maintains invoice history for financial reporting

OBJECTIVES

Project Goals and Achievements:

- ➤ Develop a full-stack web application integrating HTML, CSS, JavaScript frontend with Java Spring Boot backend
- ➤ Implement an inventory and financial management system for customers, products, sales, purchases, and expenses
- ➤ Apply data structures and algorithms concepts such as searching, sorting, and list management in real-world applications
- ➤ Create a modular and scalable architecture for easy system maintenance and feature expansion
- ➤ Build RESTful API communication between frontend dashboard and backend services
- ➤ Design responsive and user-friendly interfaces following modern web development standards
- ➤ Provide a centralized dashboard for monitoring business transactions and financial performance

COs ATTAINMENTS – FWD

Full-Stack Web Development Course Outcomes:

- **CO1 – HTML Semantic Structure: Implemented semantic tags such as header, nav, section, article, footer, and form elements across multiple pages including signup, dashboard, customers, products, sales, and invoices**
- **CO2 – Advanced CSS Styling: Applied custom styling, Flexbox layouts, responsive design, gradients, transitions, and media queries to create a modern and responsive dashboard interface**
- **CO3 – JavaScript Fundamentals: Utilized variables, operators, loops (forEach), conditional statements, objects, arrays, and functions for implementing business logic across system modules**
- **CO4 – DOM & Async Operations: Implemented DOM manipulation, event handling, Fetch API with async/await, localStorage data management, and dynamic table updates for interactive user operations**
- **CO5 – Modern Web Standards: Applied ES6 features, modular JavaScript structure, responsive UI design, and error handling techniques across the entire application to ensure maintainability and scalability**

COs ATTAINMENTS – DSA

Data Structures & Algorithms Course Outcomes:

- ➤ **CO1 – Algorithm Analysis:** Applied Big-O notation concepts by analyzing linear O(n) operations such as array traversal and data aggregation, and efficient hash-based lookups using LocalStorage
- ➤ **CO2 – Searching & Sorting:** Implemented linear search and sorting techniques to manage and display collections such as products, invoices, and transactions dynamically in the dashboard
- ➤ **CO3 – ADTs & Lists:** Utilized arrays and list-based data structures to store and manage customers, products, sales, purchases, and invoice records within the application
- ➤ **CO4 – Hashing & Data Storage:** Implemented key-value data storage using LocalStorage, similar to HashMap behavior for quick data retrieval and management
- ➤ **CO5 – Real-World Application:** Applied academic data structures and algorithm concepts to develop a practical business management dashboard integrating frontend and backend operations

Professional Invoice Dashboard

Manage your business finances with ease

Dashboard | Customers | Products | Sales | Purchases | Expenses | Invoices | Profile

50,000	30,000	15,000	5,000
Total Sales	Total Purchases	Total Expenses	Net Profit

Professional Invoice Dashboard

Manage your business finances with ease

Dashboard | Customers | Products | Sales | Purchases | Expenses | Invoices | Profile

Customers Management

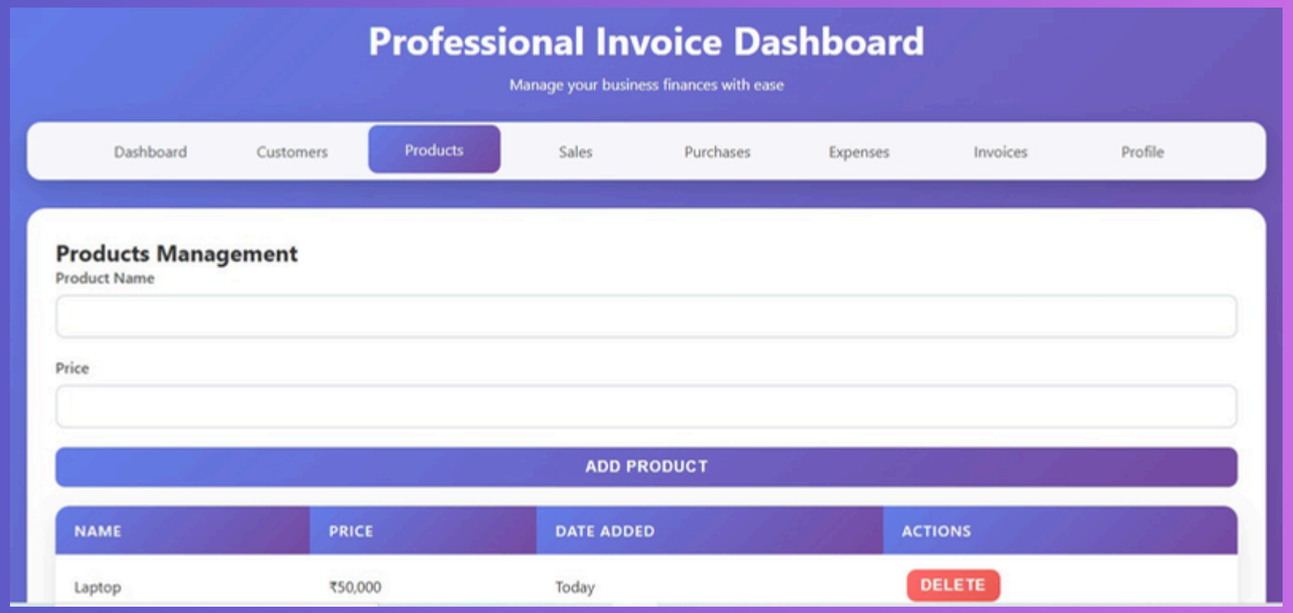
Customer Name

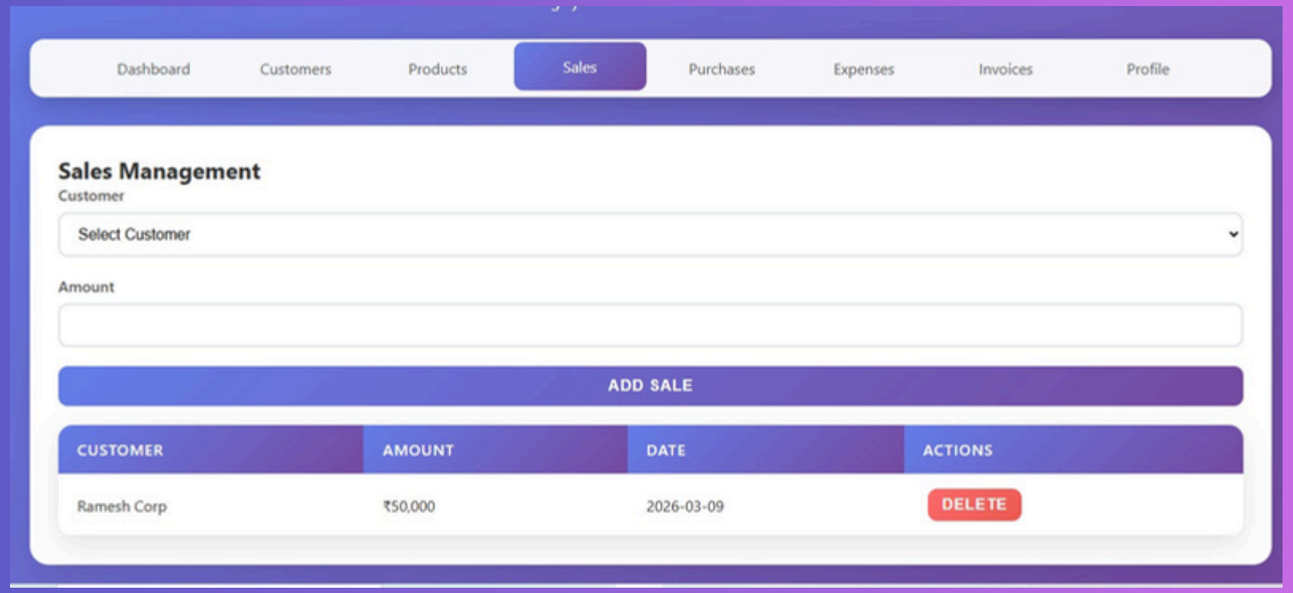
Email

ADD CUSTOMER

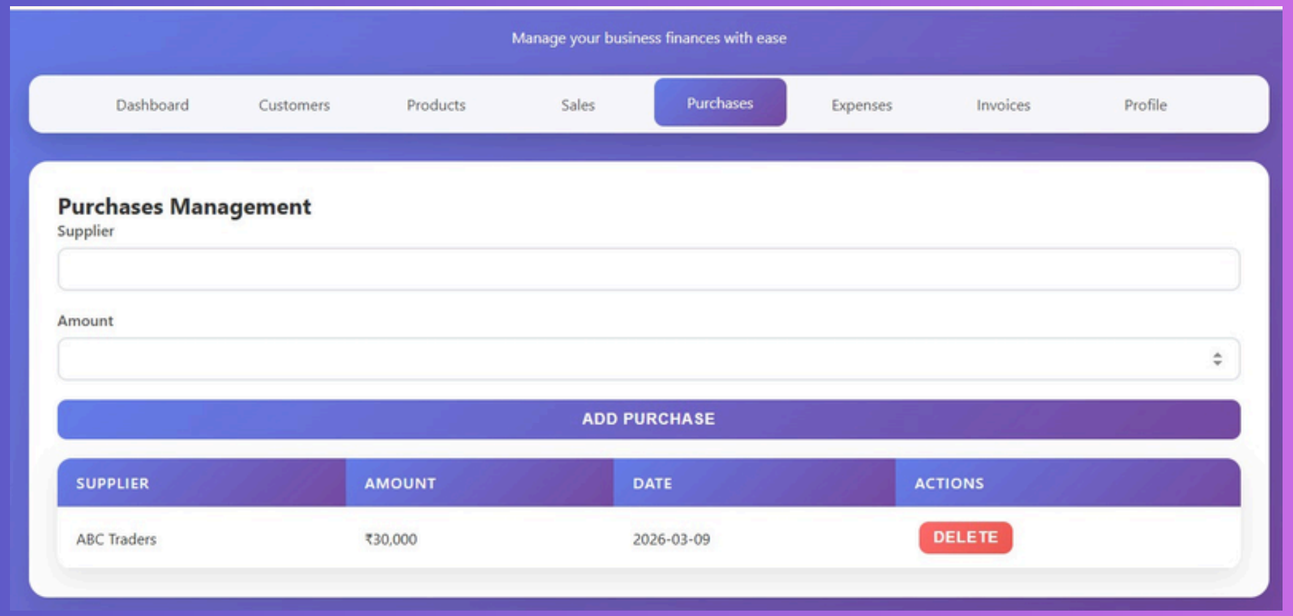
NAME	EMAIL	DATE ADDED	ACTIONS
Ramesh Corp	ramesh@example.com	2026-03-09	DELETE

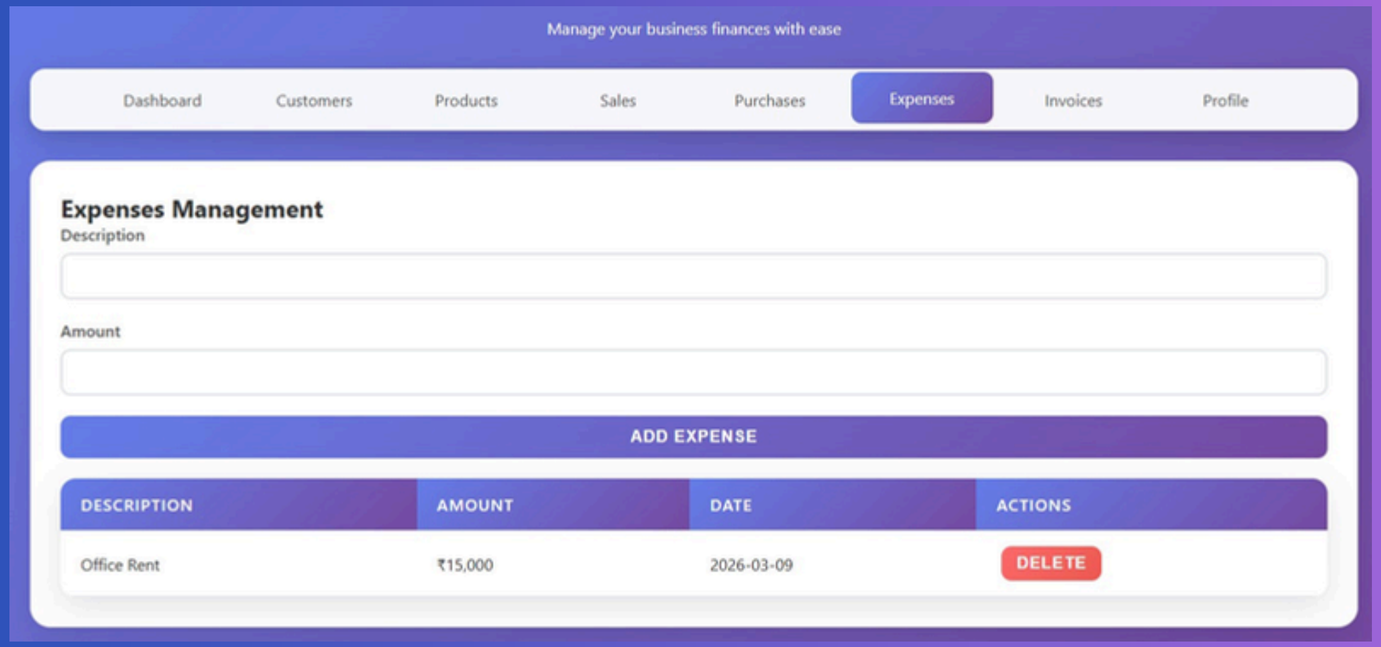
Screenshot


Professional Invoice Dashboard
Manage your business finances with ease
Dashboard
Customers
Products
Sales
Purchases
Expenses
Invoices
Profile
Products Management
Product Name
Price
ADD PRODUCT
NAME
PRICE
DATE ADDED
ACTIONS
Laptop
₹50,000
Today
DELETE

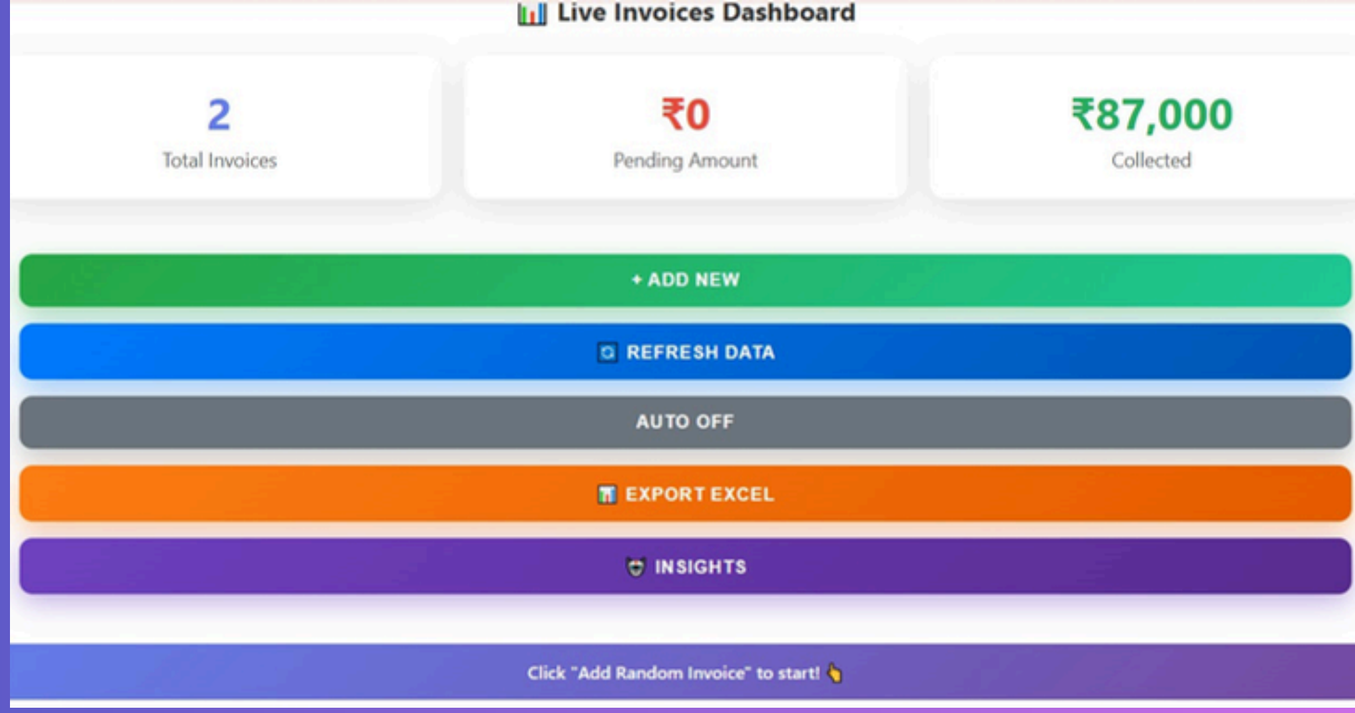

Dashboard
Customers
Products
Sales
Purchases
Expenses
Invoices
Profile
Sales Management
Customer
Select Customer
Amount
ADD SALE
CUSTOMER
AMOUNT
DATE
ACTIONS
Ramesh Corp
₹50,000
2026-03-09
DELETE

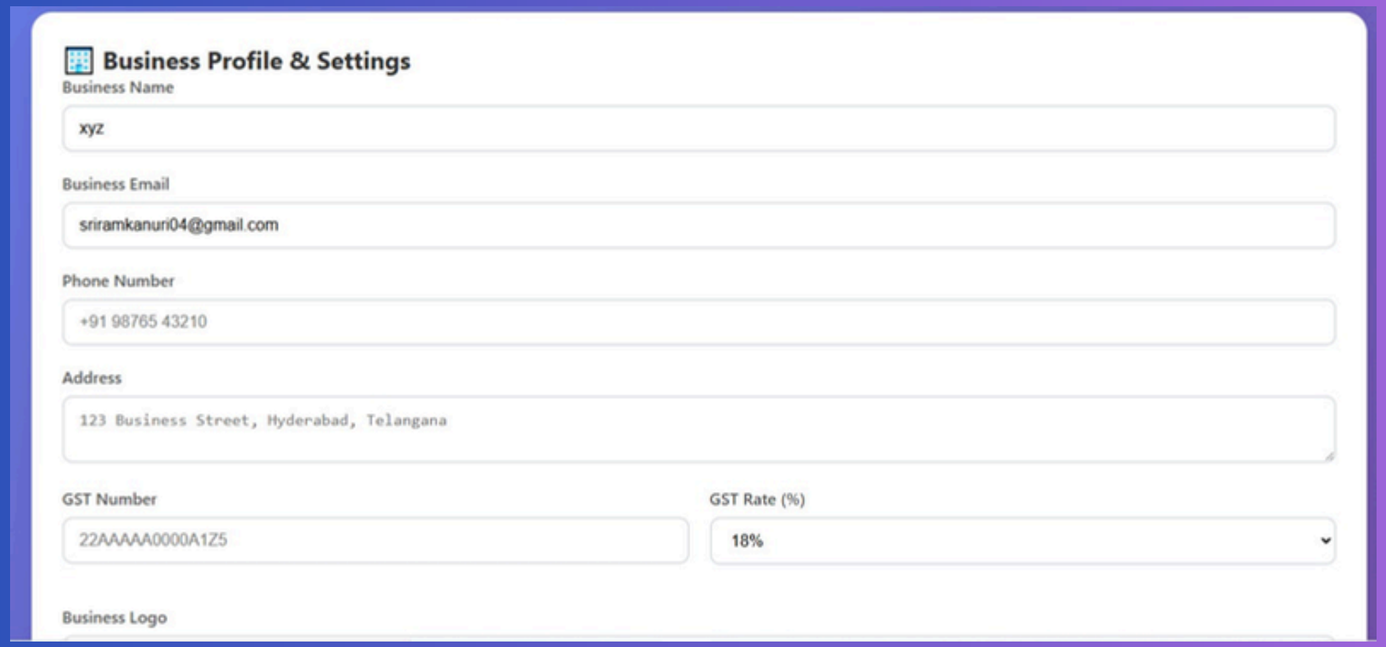
Screenshots





Scrrenshots





Conclusion

Project Outcomes and Impact:

➤ Successfully developed a full-stack business management dashboard integrating frontend and backend technologies

➤ Integrated Java Spring Boot backend with dynamic HTML, CSS, and JavaScript frontend using RESTful APIs

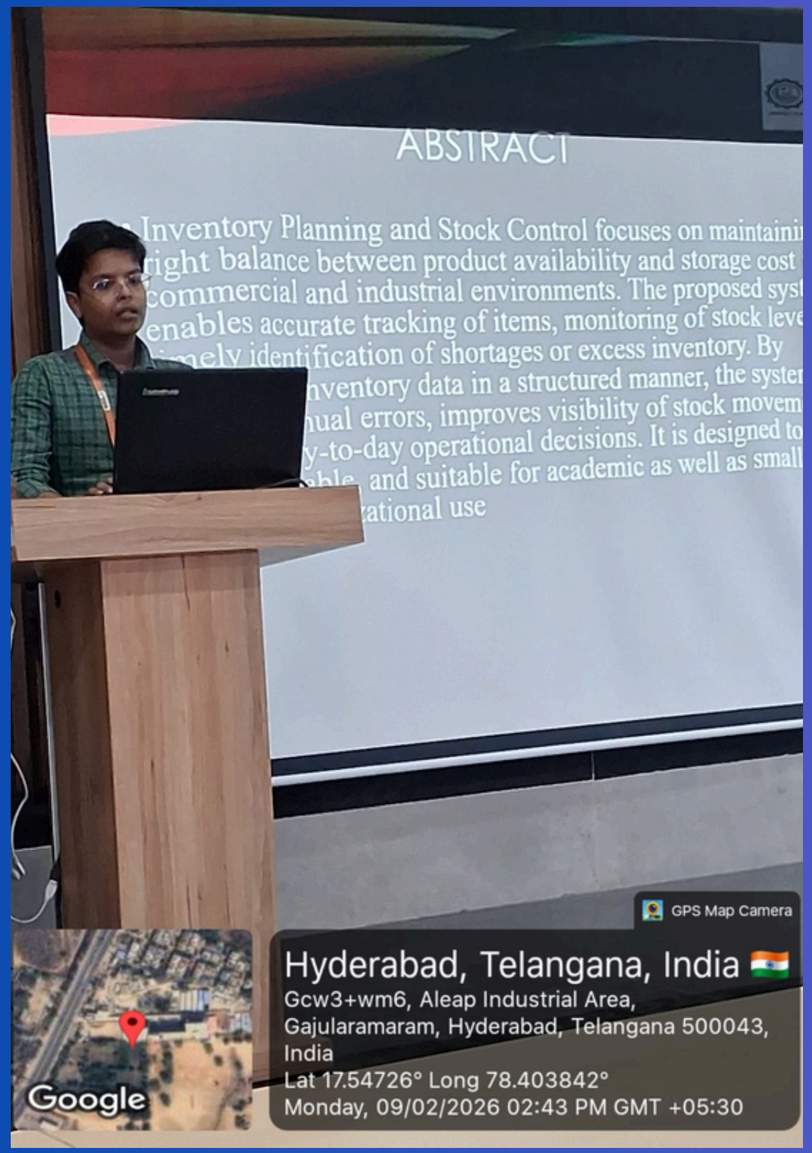
➤ Implemented practical DSA concepts such as searching, sorting, arrays, and hashing for efficient data handling

➤ Achieved comprehensive course outcome coverage across FWD and DSA domains through real-world application

➤ Built a scalable and modular architecture supporting efficient inventory tracking and financial monitoring

➤ Demonstrated the ability to transform theoretical computer science concepts into a functional business management system

➤ Future Enhancements: Cloud database integration, mobile application support, advanced analytics, and AI-based business insights


ABSTRACT
Inventory Planning and Stock Control focuses on maintaini
right balance between product availability and storage cost
commercial and industrial environments. The proposed syst
enables accurate tracking of items, monitoring of stock leve
timely identification of shortages or excess inventory. By
nventory data in a structured manner, the syster
nual errors, improves visibility of stock movem
y-to-day operational decisions. It is designed to
ble, and suitable for academic as well as small
rational use
GPS Map Camera
Hyderabad, Telangana, India
Gcw3+wm6, Aleap Industrial Area,
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Monday, 09/02/2026 02:43 PM GMT +05:30
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Thank you!

Sriram kanuri